

Biotechnology & Bioethics: Careers & Concerns in Science

Mr. Queenan – Biology 9

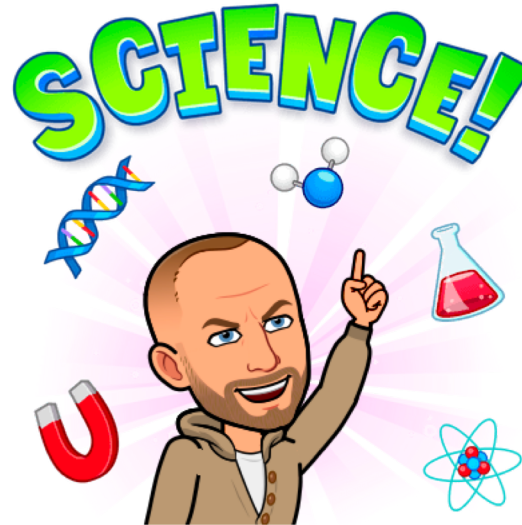
Unit 1.5 – text Ch 1.3 (p 26-29) & Ch 16.4 (p 524-529)

The Biologist

- What is a biologist?
- Are all biologists scientists?
- Is education necessary to be a biologist?
- Why do scientists, especially biologists, feel an ethical responsibility to the world around them?

The Biologist

- Biologist
 - an expert in or student of the branch of science concerning living organisms.
- Scientist
 - a person who is studying or has expert knowledge of one or more of the natural or physical sciences.
- The difference between a biologist and a biological scientist is research.



Careers in Biology

- What can you do with a biology degree?
 - Research Scientist
 - Physician
 - Pharmacologist
 - Ecologist
 - Nature Conservation Officer
 - Biotechnologist
 - Forensic Scientist
 - Government Agency and Policy Roles
 - Science Writer
 - Bioengineer
 - Patent Attorney
 - Teacher
 - More, and more, and more

Biology Through History

- **Biology through history**

- Did biology exist before we started studying it as a field in schools?

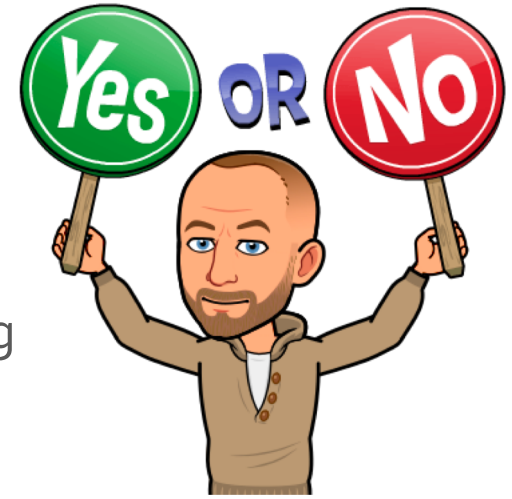
- YES!

- Did biotechnology exist prior to the development of sophisticated technology in the 20th century?

- YES!

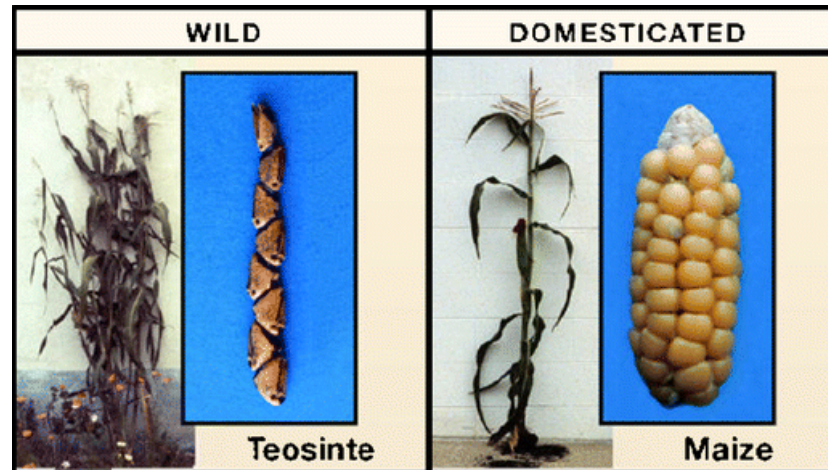
- Has our knowledge of the world around us made us more ethical or less ethical in the scope of biology?

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Evolution Connection & Biotechnology

- Evolution is a core theme of biology.
- Humans selectively breed plants and animals in the process of **artificial selection** to produce
 - more productive crops,
 - better livestock, and
 - a great variety of pets that bear little resemblance to their wild ancestors.



Biotechnology

- **Biotechnology** is simply defined as the exploitation of biological processes for industrial and other purposes
- Humans also unintentionally caused
 - the evolution of antibiotic-resistant bacteria
 - the evolution of pesticide-resistant pests
 - the loss of species through habitat loss and global climate change.
- Could this have been avoided?

Connections

- Biology, technology, and society are connected in important ways
- Many issues facing society
 - are related to biology and often involve our expanding technology.
- The basic goals of science and technology differ.
 - The goal of science is to understand natural phenomena.
 - The goal of **technology** is to apply scientific knowledge for some specific purpose.
- Although their goals differ, science and technology are interdependent.
 - Research benefits from new technologies.
 - Technological advances stem from scientific research.
- Ex. Technologies of DNA manipulation are the results of scientific discovery of the structure of DNA.
- What are other examples?

Bioethics

- Breakout questions

1. What do you think is meant by Bioethics?
2. Is it possible to be an ethical scientist?
3. What are factors that could stand in the way of ethical considerations by individuals?
4. What are examples of individuals and/or groups in society looking out for ethics in scientific research?

